

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur



Software Engineering

Assignment-0

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

A software process model represents which of the following?

a) The steps through which software is developed
b) The artifacts that are produced during software development
c) How software is used by the different categories of users
d) How software may fail and the maintenance steps for the same
e) How can the time and space complexity of the software be improved

Correct Answer: a. The steps through which software is developed,
b. The artifacts that are produced during software development

Detailed Solution:

A software process model depicts the steps through which software is developed and the artifacts that are produced during software development

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QUESTION 2:

Which of the following types of testing are conducted during unit testing?

a. Black-box testing
b. Performance testing
c. White-box testing
d. GUI testing
e. Stress testing

Correct Answer: a. Black-box testing
c. White-box testing

Detailed Solution:

Black-box and white-box both perform during unit testing.

QUESTION 3:

Which one of the following phases accounts for the maximum effort during a typical software development process?

a. Coding
b. Specification
c. Testing
d. Designing
e. Maintenance

Correct Answer: **e.** Maintenance

Detailed Solution:

In a typical **software development lifecycle**, the **maintenance phase** accounts for the **maximum effort and cost**.

QUESTION 4:

Which of the following types of testing are carried out during the testing phase of the waterfall life cycle model?

a. Unit testing
b. Integration testing
c. System testing
d. Performance testing
e. White-box testing

Correct Answer: b. Integration testing
c. System testing
d. Performance testing

Detailed Solution:

Only **b, c, and d** are part of the **testing phase** in the **Waterfall model**.

QUESTION 5:

Which of the following techniques are primarily used to handle the complexity in a large software development project?

a. Abstraction
b. Decomposition
c. Testing
d. Maintenance
e. Review

Correct Answer: a. Abstraction,
b. Decomposition

Detailed Solution:

In large software development projects, **managing complexity** is crucial. The two **primary techniques** used for this purpose are: Abstraction and Decomposition.

QUESTION 6:

Defects get introduced into a work product due to mistakes committed by the members of the development team. A mistake in an activity pertaining to which of the following phases is likely to be most expensive to correct when detected during system testing?

a. System testing
b. Design
c. Coding and unit testing
d. Requirements analysis and specification
e. Maintenance

Correct Answer: d. Requirements analysis and specification

Detailed Solution:

The correct answer is Requirements analysis and specification.

QUESTION 7:

Which one of the following phrases would most appropriately complete the sentence: "During the software development life cycle, the later a defect is found, _____ ."

a. the easier it is to find
b. the more expensive it is to repair
c. the less important it is to the product
d. the faster it is possible to fix
e. the more trivial it is

Correct Answer: b. the more expensive it is to repair

Detailed Solution:

In the software development life cycle, **defect repair cost increases significantly** the later it is detected, especially in later phases like system testing or maintenance.

QUESTION 8:

For a typical commercial software product, testing should account for what percentage of the total software development costs?

a. 5-10
b. 10-20
c. 40-50
d. 80-90
e. 90-100

Correct Answer: c. 40-50

Detailed Solution:

For a typical commercial software product, **testing often accounts for 40–50%** of the total development costs, reflecting its critical role in ensuring quality and reliability.

QUESTION 9:

Which among the following are software service type of projects?

a. Undertaking maintenance of a legacy software for a certain customer
b. Development of an off-the-shelf telemedicine software for use in hospitals across the world
c. Undertaking testing of a large software which has been outsourced by a software development organization
d. Development of a commercial relational database management system
e. Development of an embedded safety-critical application

Correct Answer: a. Undertaking maintenance of a legacy software for a certain customer
c. Undertaking testing of a large software which has been outsourced by a software development organization

Detailed Solution:

Software service type of projects are those where a company provides **services to clients** rather than developing generic or commercial products. These typically include activities like **maintenance, testing, customization, or support** for client-specific needs.

QUESTION 10:

Which of the following problems would be encountered when a programmer resorts to indiscriminate use of GO TO statements while writing a program?

a. It increases the running time of programs
b. It increases memory requirements of programs
c. It results in larger executable code sizes
d. It would make testing the code very difficult
e. It would make debugging difficult

Correct Answer: d. It would make testing the code very difficult
e. It would make debugging difficult

Detailed Solution:

Indiscriminate use of **GO TO statements** leads to **"spaghetti code"**, which lacks clear structure and flow, making it hard to follow, test, and maintain.

**********END**********